RPI Wi-Fi Dataset README

Version: 1

1: Dataset Overview

Title: RPI Wi-Fi

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Course: RPI Data Science

Purpose: To see if wifi performance varies between buildings, within buildings, and over time on RPI campus to allow students to best identify what buildings to visit for internet intensive activities.

Coverage:

- Collection Dates:
  - 10/11/2026-10/13/2026
- Locations:
  - Carnigee, Pittsburgh, Sage, Amos_Eaton, Folsom_Library, Muller, Lally, Vorhees_Computer_Center, DCC, Union, MILL_Low, Low, JEC, Academy_Hall

Reading Count: 63 after filter

2: File and Record Structure

Each row in final.csv represents one Wi-Fi test. Measurements were collected using fast.com

Files included: Raw data files, filtered data files, final.csv, main.R, graphs, README

Record Identifier: To be added

File Conventions: Using ‘,’ as CSV delimiter. MM/DD/YYYY.

3: Collection Procedure

Devices: Mobile phones
Networks: RPIWPA2
Sampling: Picked random rooms in a building we could access. Sit at a random table in the room and start the test.
Collection: Once the test is completed, measurements on the site were recorded in the raw_data excel sheet made by that student for that day of measurements.

4: Data Directory

Column Name	Data Format	Definition/Details
RCSID	Text	Student ID of collector
Start_Time	Time Text	Start time of the wifi test
End_Time	Time Text	End time of the wifi test
Location	Text	Room and building the test occurred in
Date	Date Text	Date the test occured
Download_Speed_Mbps	Numeric/ mbps	Download speed in megabits per second
Upload_Speed_Mbps	Numeric/ mbps	Upload speed in megabits per second
People_In_Room	Numeric/ integer	Observed Occupancy Estimated
Class_In_Session	Boolean	Boolean identifying if a class was in session (collector discretion)
Loaded_Latency_ms	Numeric/ ms	Latency reported during network load
Unloaded_Latency_ms	Numeric/ ms	Latency reported during no network load
Notes	Text	Optional notes about the observation

5: Cleaning and Quality Checks

Once data has been recorded in the Raw_Data directory in an xlsx file, members of the team will then “filter” the data to be put into a Filtered_Data directory with a new xlsx sheet. This filtering process changes all fields to be in the standard format (such as Darian -> DCC) and accounts for any notes left by the record, such as “this recording was completed on 4G, disregard.” Records would be excluded if noted; however, if not excluded, the data would not be edited other than changing it into the correct format.

Once data has been filtered, we then combine it in the Final_CSV folder in a csv for data analysis.

6: Analysis and reproduction

Exploratory Analysis was completed in R by plotting graphs for individual attributes as well as the comparison between different attributes.

Images are stored in the Images directory and can be replicated with the included R script labeled main.R by running it in order of the file.

7: Limitations

The team faced limitations in regards to data collection as both students were busy, unable to access all buildings, and didn't want to interrupt classes to take measurements, leading to an incomplete dataset for the entire school biased to after class hours with a small sample size.

8: Ownership, access, preservation

The repository is owned by a student's personal email and will remain after Data Science F2026 ends.

The data can be accessed on this repository by the public as required for future use.

The data will be preserved on github as the student used a personal email and will not lose access after graduation unless both students decide to take it down.